

# Xianghang (Scott) Mi

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## Research Interests

Agent Security   Trustworthy AI   Cybersecurity

## Research Overview

The overarching goal of my research is to **instill trust into AI and agentic systems, and into the digital environments in which they operate**. My recent work focuses on understanding, evaluating, and hardening the security of LLMs, AI agents, and intelligent infrastructures. To achieve this, my research is organized along two complementary directions: (1) systematically uncovering new attack surfaces, abuse paths, and failure modes in AI-enabled systems, and (2) designing robust systems and protocols that provide stronger guarantees for security, privacy, and resilience. In particular, I combine data-driven security measurement, ethical attack experiments, and AI-assisted defense techniques to identify and characterize risks spanning model interaction, tool use, and networked deployment pipelines. Building on these insights, I leverage machine learning and cryptographic primitives to prototype practical defenses that improve the confidentiality, integrity, privacy, and abuse resistance of real-world AI systems.

## Education

**2015/08 – 2019/06: Indiana University, Bloomington, IN**

Ph.D. Degree, Department of Computer Science

Advisors: Professor XiaoFeng Wang and Professor Feng Qian

Thesis: Characterizing Emerging Cybersecurity Threats: An Ecosystem Approach

\* Thesis defense completed in June 2020 due to job enrollment and the COVID-19 pandemic.

**2009/09 – 2013/06: Beijing Institute of Technology, Beijing, China**

B.S. Degree, Department of Software Engineering

## Work Experience

**12/2025 – Now: Monash University**

Assistant Professor (Lecturer)

Department of Software Systems &amp; Cybersecurity

**09/2021 – 12/2025: USTC (University of Science and Technology of China)**

Pre-Tenure Full Professor

School of Computer Science and Technology

**01/2021 – 08/2021: University at Buffalo**

Tenure-Track Assistant Professor

Department of Computer Science and Engineering

**06/2019 – 01/2021: Meta, Inc**

Research Scientist in Security Infrastructure

Project: Distributed fuzzing platform for automatic vulnerability discovery.

Project: Anti-crawling and bot detection.

**05/2014 – 06/2015: Baidu, Inc**

Senior Software Engineer in Location-Based Services

Project: Developing the mobile website of Baidu's group buying service.

## Honors and Awards

**[2025]** AsiaCCS 2025 Outstanding PC Member Award**[2022]** The Education Grant from the Foundation of Yang YuanQing**[2019] Best Paper Award (3 out of 80) in CSAW Applied Research Competition:**This award is given to our paper ([SP'19-b](#)) studying the security of virtual personal assistants (i.e., AI agents).**[2019] NDSS Distinguished Paper Award (4 out of 89):** This award is given to the paper ([NDSS'19](#)) studying the take-down practices of malicious domains.

## Datasets and Tools

The following datasets and tools have been released as part of my research projects and are publicly available to the research community.

## Datasets

- **RESIP**: Datasets of residential-proxy IPs, comprising millions of distinct IPs observed to serve as residential proxies.
- **ResiFlow**: Dataset of residential-proxy traffic flows, containing over 3TB of traffic data from deployed proxy nodes.
- **SpamHunter**: The largest publicly available dataset of tens of thousands of SMS spam messages.
- **SearchIPT**: Dataset containing over 11 million illicit promotional texts (IPTs) discovered on major search engines.
- **xPIP**: Dataset comprising 12 million distinct posts of illicit promotion (PIPs) published from 580K X accounts.
- **iFTTT**: Comprehensive dataset of services and applets crawled from the iFTTT trigger-action workflow platform.

## Tools

- **The RESIP Infiltrator**: Tool for capturing residential-proxy IP addresses from back-connect proxy services.
- **The SpamHunter**: System for continuously discovering SMS spam messages reported by victims on social networks.
- **The IPT Toolchain**: Comprehensive toolchain for capturing and analyzing illicit promotional texts, including hunter, analyzer, and infiltrator modules.
- **The PIP Hunter**: Tool for capturing posts of illicit promotion on social networks, well-tested for the X platform and adaptable to other platforms.

All datasets and tools are available through their respective project websites and GitHub repositories as referenced in [my homepage](#).

## Selected Publications

**Summary:** My research has led to tens of publications in well-recognized computer security venues, e.g., IEEE Security & Privacy, USENIX Security, CCS, NDSS, IMC, CoNEXT, DSN, ACSAC, AsiaCCS, etc. Most publications can be grouped by the following research topics:

- The security of AI agents (e.g., voice assistants): [\[IMC'17\]](#) [\[SP'19-b\]](#) [\[AsiaCCS'24\]](#) [\[WWW'26\]](#)

- Network proxies and anonymity networks: [SP'19-a] [NDSS'21] [CCS'22-a] [EuroSP'25] [arXiv'24-a] [IFIP SEC'26] .
- The security of decentralized network infrastructures: [IWQoS'23] [DSN'24] [ACSAC'24] [ACISP'26]
- Privacy-preserving prevention of harmful content (e.g., spam and malware): [CCS'22-b] [arXiv'24-e] [arXiv'24-f] .
- Detecting and understanding online abuse and cybercrime: [SP'17] [NDSS'18] [NDSS'19] [Security'19] [arXiv'24-c] [WWW'25]

\* denotes co-first authors, † stands for corresponding authors.

[1] [WWW'26] **ICL-EVADER: Zero-Query Black-Box Evasion Attacks on In-Context Learning and Their Defenses** [The ACM Web Conference 2026](#)

Ningyuan He\*, Ronghong Huang\*, Qianqian Tang, Hongyu Wang, **Xianghang Mi**<sup>†</sup>, Shanqing Guo

[PDF](#) [Code](#)

[2] [ACISP'26] **Okara: Detection and Attribution of TLS Man-in-the-Middle Vulnerabilities in Android Apps with Foundation Models** [Australasian Conference on Information Security and Privacy \(ACISP\) 2026](#)

Haoyun Yang, Ronghong Huang, Yong Fang, Beizeng Zhang, Junpu Guo, Zhanyu Wu, **Xianghang Mi**<sup>†</sup>

[PDF](#)

[3] [IFIP SEC'26] **Pepper: High-bandwidth and Scalable Anonymous Broadcast with Cryptographic Privacy** [IFIP SEC Conference 2026](#)

Chenghao Li, **Xianghang Mi**<sup>†</sup>

[4] [WWW'25] **Detecting and Understanding the Promotion of Illicit Goods and Services on Twitter**

Hongyu Wang\*, Ying Li\*, Ronghong Huang, **Xianghang Mi**<sup>†</sup>.

WWW 2025.

Acceptance Rate: 20% = 409 / 2062.

[5] [EuroSP'25] **Port Forwarding Services Are Forwarding Security Risks**

Haoyuan Wang, Yue Xue, Xuan Feng, Chao Zhou, **Xianghang Mi**<sup>†</sup>.

IEEE European Symposium on Security and Privacy 2025.

[6] [arXiv'24-a] **Shining Light into the Tunnel: Understanding and Classifying Network Traffic of Residential Proxies**

Ronghong Huang, Dongfang Zhao, **Xianghang Mi**<sup>†</sup>, Jingmiao Zhang, XiaoFeng Wang.

Under submission and in arXiv preprint 2024.

**[7] [arXiv'24-b] Reflected Search Poisoning for Illicit Promotion**

*Sangyi Wu, Jialong Xue, Shaoxuan Zhou, Xianghang Mi<sup>†</sup>.*

Under submission and in arXiv preprint 2024.

**[8] [arXiv'24-c] Fostering Cyber Threat Detection Through Federated Learning**

*Yu Bi, Yekai Li, Xuan Feng, Xianghang Mi<sup>†</sup>.*

Under submission and in arXiv preprint 2024.

**[9] [arXiv'24-d] SpamDam: Towards Privacy-Preserving and Adversary-Resistant SMS Spam Detection**

*Yekai Li, Rufan Zhang, Wenxin Rong, Xianghang Mi<sup>†</sup>.*

Under submission and in arXiv preprint 2024.

**[10] [ACSAC'24] Dissecting Open Edge Computing Platforms: Ecosystem, Usage, and Security Risks**

*Yu Bi\*, Mingshuo Yang\*, Yong Fang, Xianghang Mi<sup>†</sup>, Shanqing Guo<sup>†</sup>, Shujun Tang, Haixin Duan.*

ACSAC 2024.

Acceptance Rate: 20% = 83/421.

**[11] [AsiaCCS'24] Command Hijacking on Voice-Controlled IoT in Amazon Alexa Platform**

*Wenbo Ding, Song Liao, Long Cheng, Xianghang Mi, Ziming Zhao, Hongxin Hu.*

The 19th ACM Asia Conference on Computer and Communications Security (AsiaCCS 2024).

Acceptance Rate: 18% = 418/2,322.

**[12] [DSN'24] Stealthy Peers: Understanding Security Risks of WebRTC-based Peer-Assisted Video Streaming**

*Siyuan Tang, Eihal Alowaisheq, Xianghang Mi<sup>†</sup>, Yi Chen, XiaoFeng Wang, Yanzhi Dou.*

IEEE/IFIP DSN 2024.

Acceptance Rate: 20% = 42/203.

**[13] [IWQoS'23] An Empirical Study of Storj DCS: Ecosystem, Performance, and Security**

*Hao Li, Xianghang Mi<sup>†</sup>, Yanzhi Dou, Shanqing Guo<sup>†</sup>*

IEEE/ACM IWQoS 2023.

Acceptance Rate: 23.5% = 62/264.

**[14] [CCS'22-a] An Extensive Study of Residential Proxies in China**

*Mingshuo Yang\*, Yunnan Yu\*, Xianghang Mi<sup>†</sup>, Shujun Tang, Shanqing Guo<sup>†</sup>, Yilin Li, Xiaofeng Zheng, Haixin Duan.*

ACM CCS 2022.

Acceptance Rate: 22.5% = 218/971.

**[15] [CCS'22-b] Clues in Tweets: Twitter-Guided Discovery and Analysis of SMS Spam**

*Siyuan Tang, Xianghang Mi<sup>†</sup>, Ying Li, Xiaofeng Wang, Kai Chen.*

ACM CCS 2022.

Acceptance Rate: 22.5% = 218/971.

**[16] [NDSS'21] Your Mobile Phone is My Proxy: Understanding and Detecting Mobile Proxy Networks**

*Xianghang Mi<sup>†</sup>, Siyuan Tang, Zhengyi Li, Xiaojing Liao, Feng Qian, Xiaofeng Wang.*

Network and Distributed System Security Symposium 2021.

Acceptance Rate: 15.2% = 87/573.

**[17] [SP'19-a] Resident Evil: Understanding Residential IP Proxy as a Dark Service**

*Xianghang Mi, Xuan Feng, Xiaojing Liao, Baojun Liu, Xiaofeng Wang, Feng Qian, Zhou Li, Sumayah Alrwais, Limin Sun, Ying Liu.*

IEEE Symposium on Security and Privacy 2019.

Acceptance Rate: 12% = 84/679.

**[18] [SP'19-b] Dangerous Skills: Understanding and Mitigating Security Risks of Voice-Controlled Third-Party Functions on Virtual Personal Assistant Systems**

**3rd Place (3 out of 80) of CSAW'19 Applied Research Competition**

*Nan Zhang, Xianghang Mi, Xuan Feng, XiaoFeng Wang, Yuan Tian, Feng Qian.*

IEEE Symposium on Security and Privacy 2019.

Acceptance Rate: 12% = 84/679.

**[19] [Security'19] Understanding iOS-based Crowdturfing through Hidden UI Analysis**

*Yeonjoon Lee\*, Xueqiang Wang\*, Kwangwuk Lee, Xiaojing Liao, XiaoFeng Wang, Tongxin Li, Xianghang Mi.*

USENIX Security Symposium (Security), 2019.

Acceptance Rate: 16% = 113/697.

**[20] [NDSS'19] Cracking the Wall of Confinement: Understanding and Analyzing Malicious Domain Take-downs**

**Distinguished Paper Award (4 out of 89)**

*Eihal Alowaisheq, Peng Wang, Sumayah A Alrwais, Xiaojing Liao, XiaoFeng Wang, Tasneem Alowaisheq, Xianghang Mi, Siyuan Tang, Baojun Liu.*

Network and Distributed System Security Symposium 2019, San Diego, CA.

Acceptance Rate: 17% = 89/521.

**[21] [NDSS'18] Game of Missuggestions: Semantic Analysis of Search-Autocomplete Manipulations**

*Peng Wang, Xianghang Mi, Xiaojing Liao, XiaoFeng Wang, Kan Yuan, Feng Qian, and*

*Raheem Beyah.*

Network and Distributed System Security Symposium 2018, San Diego, CA.

Acceptance Rate: 21.5% = 71/331.

**[22] [Security'17] Picking Up My Tab: Understanding and Mitigating Synchronized Token Lifting and Spending in Mobile Payment**

*Xiaolong Bai\*, Zhe Zhou\*, XiaoFeng Wang, Zhou Li, Xianghang Mi, Nan Zhang, Tongxin Li, S. Hu, Kehuan, Zhang.*

USENIX Security Symposium 2017, VANCOUVER, BC, CANADA.

Acceptance Rate: 16.3% = 85/522.

**[23] [SP'17] Under the Shadow of Sunshine: Understanding and Detecting BulletProof Hosting on Legitimate Service Provider Networks**

*Sumayah Alrwais, Xiaojing Liao, Xianghang Mi, Peng Wang, XiaoFeng Wang, Feng Qian, Raheem Beyah, Damon McCoy.*

IEEE Symposium on Security and Privacy 2017, San Jose, CA.

Acceptance Rate: 13% = 60/457.

**[24] [arXiv'17] Understanding IoT Security Through the Data Crystal Ball: Where We Are Now and Where We Are Going to Be**

*Nan Zhang, Soteris Demetriou, Xianghang Mi, Wenrui Diao, Kan Yuan, Peiyuan Zong, Feng Qian, XiaoFeng Wang, Kai Chen, Yuan Tian, Carl A Gunter, Kehuan Zhang, Patrick Tague, Yue-Hsun Lin.*

In arXiv preprint 2017.

**[25] [IMC'17] An Empirical Characterization of IFTTT: Ecosystem, Usage, and Performance**

*Xianghang Mi, Feng Qian, Ying Zhang, and Xiaofeng Wang.*

ACM Internet Measurement Conference 2017, London, UK.

Acceptance Rate: 23.4% = 42/179.

**[26] [CoNEXT'16] SMig: Stream Migration Extension For HTTP/2**

*Xianghang Mi, Feng Qian, and XiaoFeng Wang.*

International Conference on emerging Networking EXperiments and Technologies 2016, Irvine, CA.

Acceptance Rate: 17.6% = 35/199.

## Teaching

**2022-2024** Introduction to Cybersecurity, USTC

**2022** The Frontier of Cybersecurity Research, Fall 2022, USTC

**2021** CSE 489/589 Modern Networking Concepts, Spring 2021, UB

**2016-2017** Guest Lectures: CSCI P438 (Computer Networks, IU, Fall 2016), CSCI P538 (Advanced Computer Networks, IU, Fall 2016/Fall 2017).

## Professional Service

**Conference and Workshop Program Committee:** AsiaCCS 2025/2026, WWW 2026, IEEE ICPADS 2023, GenoPri 2020/2021, SKM 2021.

**Reviewer for Journals:** IEEE Transaction on Mobile Computing 2018, ACM Transactions on Privacy and Security 2021/2022, IEEE Transactions on Dependable and Secure Computing 2021/2022, IEEE Security & Privacy 2021/2022.

**Reviewer for Conferences and Workshops:** IEEE Security & Privacy 2020, ACM CHI 2020, ACM CCS 2019, IEEE INFOCOM 2019, NDSS 2022/2019/2018, IEEE ICDCS 2017, ACM AsiaCCS 2016, BIGCOM 2016.

## Funding Awards

**2025** *Blackbox and Zero-Query Evasion Attacks Against LLM-based In-Context Learning*, ¥300K, CCF

**2023** *Understanding and Mitigating the Security Risks of Residential Proxies*, ¥300K, NSFC

**2022** *Fostering Cyber Threat Detection through Federated Learning*, ¥200K, MSR Asia

**2022** *The Innovation Fund for Young Investigators*, ¥90K, USTC

## Graduated Students

- Ronghong Huang, MPhil 2025  $\implies$  Software Engineer at Tencent
- Yue Xue, MPhil 2025  $\implies$  Software Engineer at Agricultural Development Bank of China
- Yong Fang, MPhil 2025  $\implies$  Software Engineer at CSNWD Group
- Yu Bi, MPhil 2024  $\implies$  Security Engineer at Meituan
- Mingshuo Yang, MPhil 2024  $\implies$  Software Engineer at Huawei

- Mingkai Li, BS 2023  $\implies$  PhD at Columbia University
- Ying Li, MPhil 2023  $\implies$  PhD at UCLA
- Hanliang Jiang, BS 2023  $\implies$  Master at UIUC
- Wenxin Rong, BS 2022  $\implies$  PhD at University of Delaware